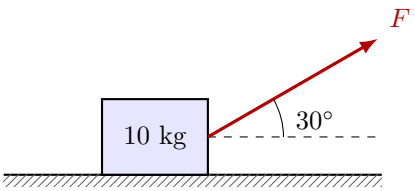
Level 3 Sheet

Time to be Taken: 80 Min Max

1. As shown in the figure a block of mass 10 kg lying on a horizontal surface is pulled by a force F acting at an angle 30°, with horizontal. For $\mu_s = 0.25$, the block will just start to move for the value of F: [Given $g = 10\ \text{m s}^{-2}$]



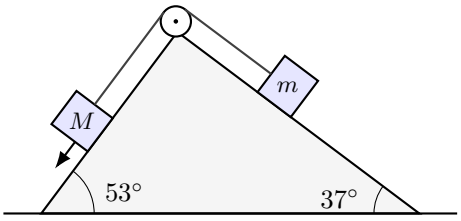
(A) 33.3 N (B) 25.2 N

(C) 20 N (D) 35.7 N

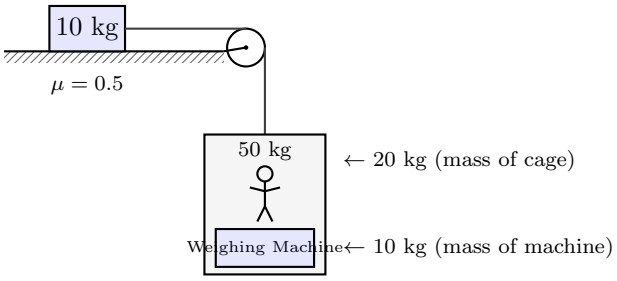
2. A spring of force constant 15 N/m is cut into two pieces. If the ratio of their length is 1 : 3, then the force constant of smaller piece is _____ N/m.

3. The length of a light string is 1.4 m when the tension on it is 5 N. If the tension increases to 7 N, the length of the string is 1.56 m. The original length of the string is _____ m.

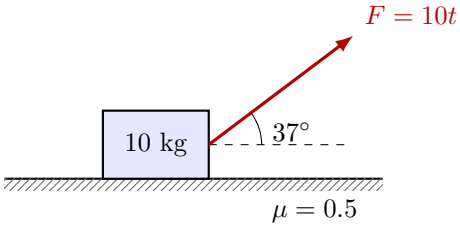
4. In the given arrangement of a doubly inclined plane two blocks of masses M and m are placed. The blocks are connected by a light string passing over an ideal pulley as shown. The coefficient of friction between the surface of the plane and the blocks is 0.25. The value of m, for which $M = 10$ kg will move down with an acceleration of 2 m s^{-2}, is: (take $g = 10\ \text{m s}^{-2}$ and $\tan 37° = \frac{3}{4}$)



5. If pulley is ideal and string is massless then reading of weighing machine is $\frac{5x}{3}$ kg then calculate x. ($g = 10\ \text{m/s}^2$)

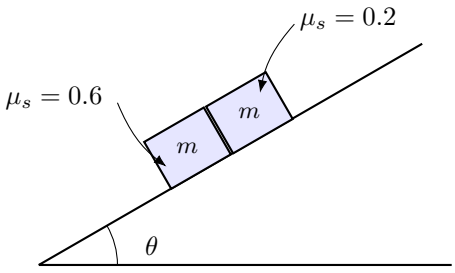


6. A time varying force $F = 10t$ (in Newton) starts acting on the block at $t = 0$. Then the time at which block will start sliding on horizontal surface is :



(1) $\frac{50}{11}$ s (2) 6.25 s (3) $\frac{100}{11}$ s (4) 5 s

7. Two blocks of equal mass are kept in touch on an inclined plane as shown in figure. If θ_0 is the minimum value of angle θ for which both the blocks start sliding then what is the value of $2\cot\theta_0$.



(1) 1 (2) 3 (3) 5 (4) 12

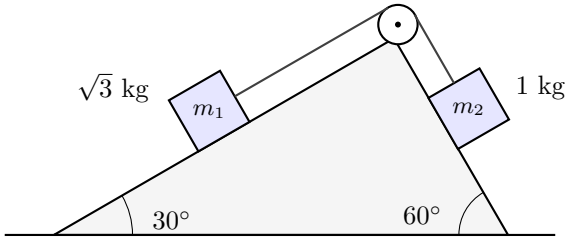
(Practice two blocks on an incline, all three cases)

8. Two blocks m_1 and m_2 are placed on a smooth inclined plane as shown in figure. If they are released from rest. Find:

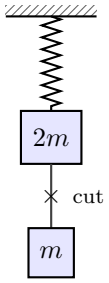
(i) acceleration of mass m_1 and m_2

(ii) tension in the string

(iii) net force on pulley exerted by string



9. System shown in figure is in equilibrium and at rest. The spring and string are massless. Now the string is cut. The acceleration of mass $2m$ and m just after the string is cut will be : **[JEE 2006, 3/184]**



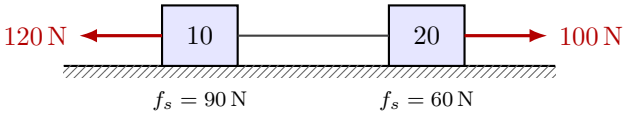
(A) $g/2$ upwards, g downwards

(B) g upwards, $g/2$ downwards

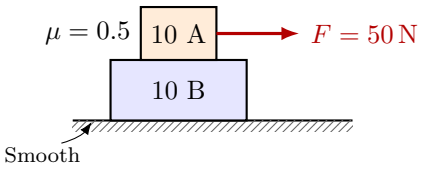
(C) g upwards, $2g$ downwards

(D) $2g$ upwards, g downwards

10. Find the tension in the string in situation as shown in the figure below. Forces 120 N and 100 N start acting when the system is at rest and the maximum value of static friction on 10 kg is 90 N and that on 20 kg is 60 N?



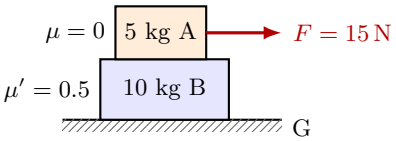
11. Find the acceleration of the two blocks. The system is initially at rest and the friction coefficient are as shown in the figure?



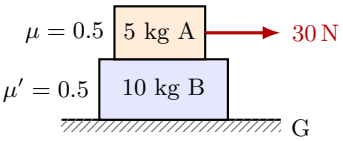
(Practice minimum 5 two block problems from notes and module combined)

12. In the given figures find the accelerations and the friction forces involved :

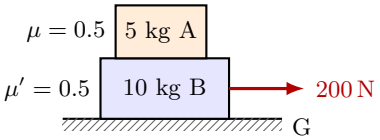
(i)



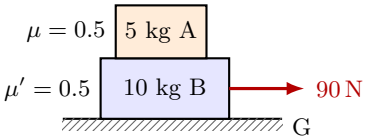
(ii)



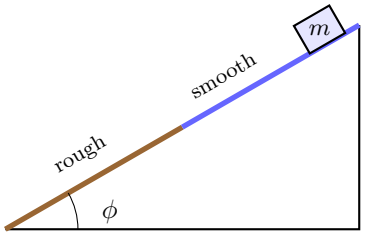
(iii)



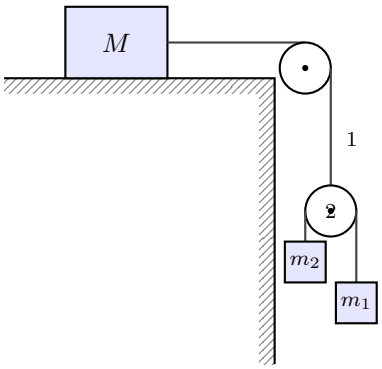
(iv)



13. The upper half of an incline plane with inclination ϕ is perfectly smooth while the lower half is rough. A body starting from rest at the top will again come to rest at the bottom if the coefficient of friction for the lower half is $x \tan\phi$ then find the value of x.



14. In the arrangement shown, $m_1 = 1\,\text{kg}$, $m_2 = 2\,\text{kg}$, the pulleys and strings are ideal and all surfaces in contact are frictionless. The value of M for which the mass m_1 moves with constant velocity is



(A) 6 kg

(B) 4 kg

(C) 8 kg

(D) 10 kg

Keep Going!

"Success is the sum of small efforts, repeated day in and day out."

— Robert Collier